

CoRoT-2b

R-1215

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-2_140928 · created 2026-07-31T08:13:13+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2456928.7195 +/- 0.0017 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.206 +/- 0.061
Transit depth (Rp/Rs)^2	4.23 +/- 2.5 [%]
Orbital Inclination (inc)	84.77 +/- 2.07
Airmass coefficient 1 (a1)	1.13 +/- 0.062
Airmass coefficient 2 (a2)	-0.068 +/- 0.015
Scatter in the residuals of the lightcurve fit is	5.06 %
Best Comparison Star	#1 - [38, 406]
Optimal Aperture	2.39
Optimal Annulus	7.24
Transit Duration (day)	0.083 +/- 0.016

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.206 $\pm$ 0.061 vs 0.1667 (deviation 0.64 σ)
Duration fit vs published	0.083 d vs 0.095 d (deviation 0.75 σ)
Mid-time O-C	-6.2 min
Depth SNR	3.4
Residual scatter	5.06 %

Light curve

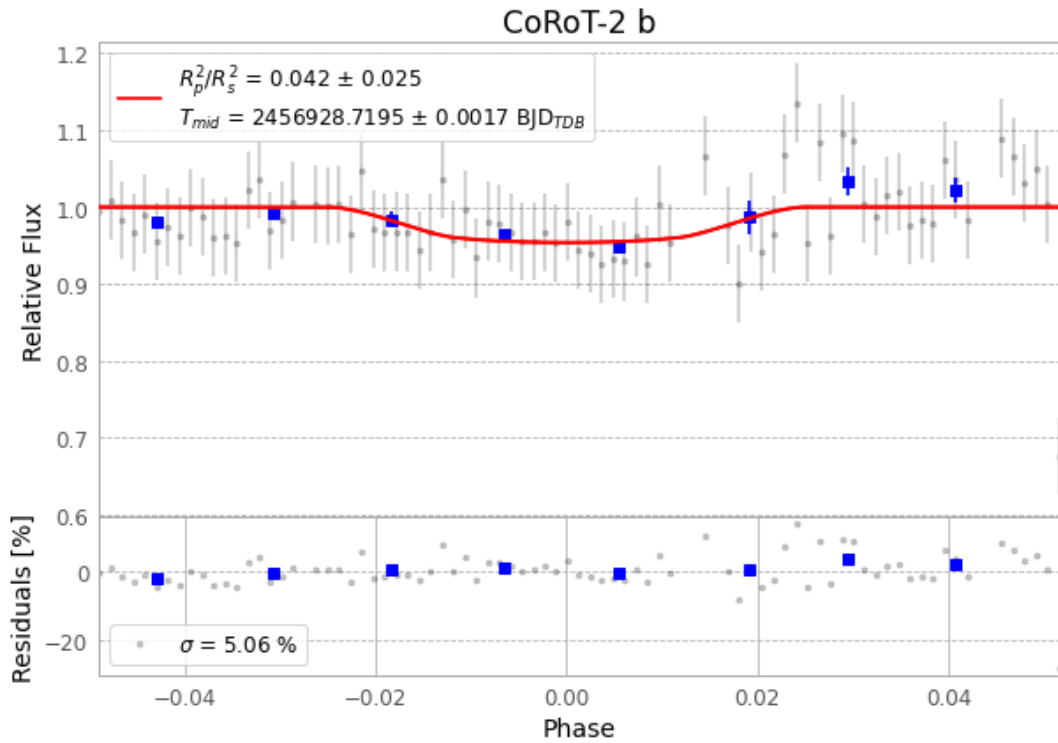


Figure 1 — FinalLightCurve_CoRoT-2 b_2014-09-27.png (SHA-256 7d755d85f649bbd7...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [242, 297], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 94ec38266e5f2b50...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

86 FITS frames (56 MB), CoRoT-2140928030913.FITS → CoRoT-2140928072413.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-140928.log (7b3535bb2025...), FinalLightCurve_CoRoT-2 b_2014-09-27.png (7d755d85f649...), FinalLightCurve_CoRoT-2 b_2014-09-27.csv (e205fd7f1768...)

Compute resources

Wall time 155.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 08:13Z.